

Table 1: Prominent and notable IoT use cases

Smart home automation • Smart thermostats • Smart lighting systems • Home security systems • Smart appliances	Industrial IoT (IIoT) • Predictive maintenance for machinery • Asset tracking and management • Quality control in manufacturing • Supply chain optimisation
Healthcare • Remote patient monitoring • Wearable health devices • Medication adherence monitoring • Smart prosthetics	Transportation and logistics • Vehicle tracking and fleet management • Traffic management and optimisation • Predictive maintenance for vehicles • Autonomous vehicles
Agriculture • Precision agriculture • Livestock monitoring • Soil and environmental monitoring • Automated irrigation systems	Smart cities • Smart traffic management • Waste management optimisation • Environmental monitoring • Smart street lighting
Retail • Inventory management and tracking • Customer behaviour analysis • Smart shelves and checkout systems • Personalised shopping experiences	Energy management • Smart grid technology • Energy consumption monitoring • Demand response systems • Renewable energy integration
Environmental monitoring • Wildlife tracking and conservation • Water quality monitoring • Forest fire detection and prevention • Pollution control	Smart buildings • Building automation and HVAC control • Energy-efficient lighting systems • Occupancy and space utilisation monitoring • Security and access control
Supply chain • Real-time tracking of goods in transit • Temperature and humidity monitoring for perishables • Inventory optimisation • Blockchain-based supply chain transparency	Security and surveillance • Smart cameras and video analytics • Intrusion detection systems • Fire and smoke detection • Access control systems
Smart farming • Crop and soil monitoring • Livestock tracking and health monitoring • Automated harvesting and sorting • Pest and disease management	Health and fitness • Fitness wearables • Smart scales and blood pressure monitors • Remote health consultations • Medication dispensers
Water management • Leak detection and prevention • Water quality monitoring • Irrigation control • Flood prediction and response	Manufacturing • Process automation and optimisation • Quality control and defect detection • Worker safety monitoring • Inventory tracking in real-time
Smart tourism • Location-based services and recommendations • Smart transportation options • Tourist information kiosks • Cultural heritage preservation and monitoring	Education • Smart classrooms and interactive learning • Student attendance monitoring • Campus security and access control • Asset tracking for educational institutions
Wearable technology • Smartwatches • Fitness trackers • Augmented reality (AR) glasses • Healthcare monitoring devices	Sports and entertainment • Sports performance analytics • Smart stadiums and fan engagement • Ticketing and access control • Equipment tracking and maintenance